

Paper: 1MA1/3H					
Question		Answer	Mark	Mark scheme	Additional guidance
18	(a)	$6x^3 + 35x^2 + 58x + 21$	M1	for a method to find the product of two linear expressions, 3 correct terms out of 4 terms e.g. $2x^2 + x + 6x + 3$ or $3x^2 + 7x + 9x + 21$ or $6x^2 + 14x + 3x + 7$	Note that, for example, $7x + 3$ is regarded as three terms in the expansion of $(2x + 1)(x + 3)$ First product must be a 3 or 4 term quadratic but need not be simplified or may be simplified incorrectly Accept $a = 6, b = 35, c = 58, d = 21$
			M1	for a complete method to obtain all terms, at least half of which are correct (ft their first product) e.g. $6x^3 + 32x^2 + 42x + 3x^2 + 16x + 21$	
			A1	cao	
	(b)	$\frac{2}{5} < x < 1\frac{3}{5}$	M1	for first step of finding the square root of both sides eg $1 - x < \pm \frac{3}{5}$ OR for writing in the form $ax^2 + bx + c (< 0)$ eg $x^2 - 2x + \frac{16}{25} (< 0)$ or $25x^2 - 50x + 16 (< 0)$	Condone use of an “=” sign; accept one square root (eg $\frac{3}{5}$) only shown. Critical values can be stated, or shown in an expression (which may have incorrect inequality symbols) Could be written as two separate expressions eg $x > \frac{2}{5}$ and $x < 1\frac{3}{5}$ oe
			M1	for showing critical values $\frac{2}{5}$ (= 0.4) and $1\frac{3}{5}$ (= 1.6) oe	
			A1	for $\frac{2}{5} < x < 1\frac{3}{5}$ oe	